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Summary

I work on multimodal generation, currently on turning the text interface of a generative model into something that can be measured and scaled. My earlier work covers 3D and 4D reconstruction and generation, including text-to-3D with Gaussian splatting, video diffusion models as 3D generators, and native mesh generation, published at CVPR, NeurIPS, and T-PAMI. Before Tsinghua I worked on knowledge graphs and their applications in natural language processing.

Education

PhD Tsinghua University, Computer Science (Sept 2022 – present)

- Advisor: [Huaping Liu](#)

BS Xi'an Jiaotong University, Physics (Sept 2018 – July 2022)

- Advisor: [Minnan Luo](#)

Experience

ByteDance Seed., Top Seed intern on multimodal generation, led by [Haoqi Fan](#) – Beijing, China (July 2025 – present)

- **Seedream 5.0:**
 - Designed the structured prompt (SP) annotation pipeline and verified the effectiveness of SP on DiT.
 - Trained the prompter rewriting user prompts into SPs, exploring reward models, RL recipes, self-distillation, and thinking patterns.
 - Built the editing SP data pipeline, mining supervision from video and extending structured prompts from generation to instruction-based editing.
- **Scaling Properties of Text Conditioning in Visual Generation [1]:**
 - Showed that scaling visual generation means scaling caption informativeness and the LLM that produces it, not the DiT alone.
 - Across 15 controlled runs with data, architecture, and compute fixed, converged diffusion loss tracks caption informativeness, not caption length.
 - Proposed the diffusability × promptability decomposition of the caption interface, separating what the diffuser can use from what an LLM can instantiate.
- **Open-sourced unified model and data engine:**
 - Co-led LightFusion [3], fusing off-the-shelf generation and understanding models via interleaved multimodal self-attention: 0.91 GenEval and 82.16 DPG-Bench on ~35B tokens.
 - Co-led VQ-VA World [2], an agentic pipeline crawling ~1.8M interleaved image-text samples plus the Intelligent-Bench benchmark, lifting LightFusion from 7.78 to 53.06.
- **Seed world model** (ongoing):
 - Continued training on top of Seedance 2.5.
 - Built the captioning workflow, annotating both event-level and global information.
 - Owned its prompt enhancement model, proposing an offline + online scheme that meets the world model's latency budget.
 - Designed the 4D structured prompt and reference-to-world interfaces, giving control through 3D bounding boxes, reference views, and object references.
- **Negative results:** tested a pretrained ViT as the VAE and a fully discrete unified model, among other alternatives to the continuous latent; none outperformed VAE-based latent diffusion in our settings.

Shengshu Inc., Research intern on video and 3D generation – Beijing, China (Nov 2023 – Mar 2025)

- Alleviating Janus problem in optimization-based 3D generation methods. (CVPR 2024 [6])
- Finetune video diffusion model for 3D generation.

- Native 3D generation using limited 3D data. (CVPR 2025 Highlight [4])

Publications (= Indicates Equal Contribution)

[1] **Scaling Properties of Text Conditioning in Visual Generation** – *ByteDance Seed 2026*

Zilong Chen, Chaorui Deng, Kunchang Li, Hongyi Yuan, Haoqi Fan

heheyas.github.io/context-scaling [🔗](#)

[2] **VQ-VA World: Towards High-Quality Visual Question-Visual Answering** – *CVPR 2026*

Chenhui Gou=, **Zilong Chen**=, Zeyu Wang=, Feng Li, Deyao Zhu, Zicheng Duan, Kunchang Li, Chaorui Deng, Hongyi Yuan, Haoqi Fan, Cihang Xie, Jianfei Cai, Hamid Rezaatofghi

chenhuigou.github.io/VQ-VA-World [🔗](#)

[3] **LightFusion: A Light-weighted, Double Fusion Framework for Unified Multimodal Understanding and Generation** – *ECCV 2026*

Zeyu Wang=, **Zilong Chen**=, Chenhui Gou=, Feng Li, Chaorui Deng, Deyao Zhu, Kunchang Li, Weihao Yu, Haoqin Tu, Haoqi Fan, Cihang Xie

arxiv.org/abs/2510.22946 [🔗](#)

[4] **MeshGen: Generating PBR Textured Mesh with Render-Enhanced Auto-Encoder and Generative Data Augmentation** – *CVPR 2025 (Highlight)*

Zilong Chen, Yikai Wang, Wenqiang Sun, Feng Wang, Yiwen Chen, Huaping Liu

heheyas.github.io/MeshGen [🔗](#)

[5] **V3D: Video Diffusion Models Are Effective 3D Generators** – *T-PAMI 2025*

Zilong Chen, Yikai Wang, Feng Wang, Zhengyi Wang, Huaping Liu

heheyas.github.io/V3D [🔗](#)

[6] **Text-to-3D Using Gaussian Splatting** – *CVPR 2024*

Zilong Chen, Feng Wang, Yikai Wang, Huaping Liu

gsgen3d.github.io [🔗](#)

[7] **GaussianEditor: Swift and Controllable 3D Editing with Gaussian Splatting** – *CVPR 2024*

Yiwen Chen=, **Zilong Chen**=, Chi Zhang, Feng Wang, Xiaofeng Yang, Yikai Wang, Zhongang Cai, Lei Yang, Huaping Liu, Guosheng Lin

buaacyw.github.io/gaussian-editor [🔗](#)

[8] **Masked Space-Time Hash Encoding for Efficient Dynamic Scene Reconstruction** – *NeurIPS 2023 (Spotlight)*

Feng Wang=, **Zilong Chen**=, Guokang Wang, Yafei Song, Huaping Liu

masked-spacetime-hashing.github.io [🔗](#)

[9] **Video4DGen: Enhancing Video and 4D Generation Through Mutual Optimization** – *T-PAMI 2025*

Yikai Wang, Guangce Liu, Xinzhou Wang, **Zilong Chen**, Jiafang Li, Xin Liang, Fuchun Sun, Jun Zhu

vidu4d-dgs.github.io [🔗](#)

[10] **Vidu4D: Single Generated Video to High-Fidelity 4D Reconstruction with Dynamic Gaussian Surfels** – *NeurIPS 2024*

Yikai Wang, Xinzhou Wang, **Zilong Chen**, Zhengyi Wang, Fuchun Sun, Jun Zhu

vidu4d-dgs.github.io [🔗](#)

[11] **MeshAnything V2: Artist-Created Mesh Generation with Adjacent Mesh Tokenization** – *arXiv 2024*

Yiwen Chen, Yikai Wang, Yihao Luo, Zhengyi Wang, **Zilong Chen**, Jun Zhu, Chi Zhang, Guosheng Lin

buaacyw.github.io/meshanything-v2 [🔗](#)

[12] **DimensionX: Create Any 3D and 4D Scenes from a Single Image with Controllable Video Diffusion** – *arXiv 2024*

Wenqiang Sun, Shuo Chen, Fangfu Liu, **Zilong Chen**, Yueqi Duan, Jun Zhang, Yikai Wang

chenshuo20.github.io/DimensionX [🔗](#)

[13] **FreePlane: Unlocking Free Lunch in Triplane-Based Sparse-View Reconstruction Models** – *arXiv 2024*

Wenqiang Sun, Zhengyi Wang, Shuo Chen, Yikai Wang, **Zilong Chen**, Jun Zhu, Jun Zhang
freeplane3d.github.io 

[14] TwiBot-22: Towards Graph-Based Twitter Bot Detection – *NeurIPS 2022*

Shangbin Feng=, Zhaoxuan Tan=, Herun Wan=, Ningnan Wang=, **Zilong Chen=**, Binchi Zhang=, Qinghua Zheng, Wenqian Zhang, Zhenyu Lei, Shujie Yang, others

[15] Knowledge Graph Augmented Political Perspective Detection in News Media – *arXiv 2021*

Shangbin Feng, **Zilong Chen**, Qingyao Li, Minnan Luo

[16] Encoding Heterogeneous Social and Political Context for Entity Stance Prediction – *arXiv 2021*

Shangbin Feng, **Zilong Chen**, Peisheng Yu, Minnan Luo

[17] KCD: Knowledge Walks and Textual Cues Enhanced Political Perspective Detection in News Media – *NAACL 2022 (Oral)*

Wenqian Zhang=, Shangbin Feng=, **Zilong Chen=**, Zhenyu Lei, Jundong Li, Minnan Luo

[18] BIC: Twitter Bot Detection with Text-Graph Interaction and Semantic Consistency – *ACL 2023*

Zhenyu Lei, Herun Wan, Wenqian Zhang, Shangbin Feng, **Zilong Chen**, Jundong Li, Qinghua Zheng, Minnan Luo

[19] KRACL: Contrastive Learning with Graph Context Modeling for Sparse Knowledge Graph Completion – *WWW 2023*

Zhaoxuan Tan, **Zilong Chen**, Shangbin Feng, Qingyue Zhang, Qinghua Zheng, Jundong Li, Minnan Luo

[20] KGAP: Knowledge Graph Augmented Political Perspective Detection in News Media – *arXiv 2021*

Shangbin Feng, **Zilong Chen**, Wenqian Zhang, Qingyao Li, Qinghua Zheng, Xiaojun Chang, Minnan Luo

[21] PAR: Political Actor Representation Learning with Social Context and Expert Knowledge – *EMNLP 2022*

Shangbin Feng, Zhaoxuan Tan, **Zilong Chen**, Ningnan Wang, Peisheng Yu, Qinghua Zheng, Xiaojun Chang, Minnan Luo

Awards

Huiyan Scholarship (Tsinghua University) (2024)

Track winner, BMW hackathon (2023)

Technologies

Languages: C++, C, CUDA, Python

Software: Blender